

Project 3

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- **Organization:**

- Individually work on easier bullet points in a first-come-first-served style
- Then we would discuss and implement the more challenging aspects together

- **Implementation:**

- Java 25
- JDBC for database access
- Database is local PostgreSQL server
- Used SAX to parse the XML

- In general followed the deadline schedule
- This was somewhat tight, as both of us were busy with other responsibilities, leading to some stress
- Nevertheless, worked quite well overall, since we never missed the deadline

- “Explain how you solved the recursion problem for the Edge model” - ??

Descendants:

```
WITH RECURSIVE descendants AS (  
    SELECT e.to_ AS id  
    FROM Edge AS e  
    WHERE e.from_ = %id  
    UNION ALL  
    SELECT e.to_ AS id  
    FROM Edge AS e  
    INNER JOIN descendants AS d  
    ON e.from_ = d.id  
) SELECT * FROM descendants NATURAL JOIN Node
```

Ancestors:

```
WITH RECURSIVE ancestors AS (  
    SELECT e.from_ AS id  
    FROM Edge AS e  
    WHERE e.to_ = %id  
    UNION ALL  
    SELECT e.from_ AS id  
    FROM Edge AS e  
    INNER JOIN ancestors AS d  
    ON e.to_ = d.id  
) SELECT * FROM ancestors NATURAL JOIN Node
```

- PostgreSQL 18.3, Java 25.0.2 (JDBC 42.0.6)
- Client & Server on same machine
- **Machine:**
 - **CPU:** Ryzen 5 5600H 6-Core 12-Threads 3.3 Ghz
 - **RAM:** 16GB, 3200 MHz DDR4
 - **OS:** Fedora Linux 44
- Parameter is the size of the dataset, simulated by the number n of venues we consider ($n \in \{3, 6, 12, 24\}$)
- 100 runs

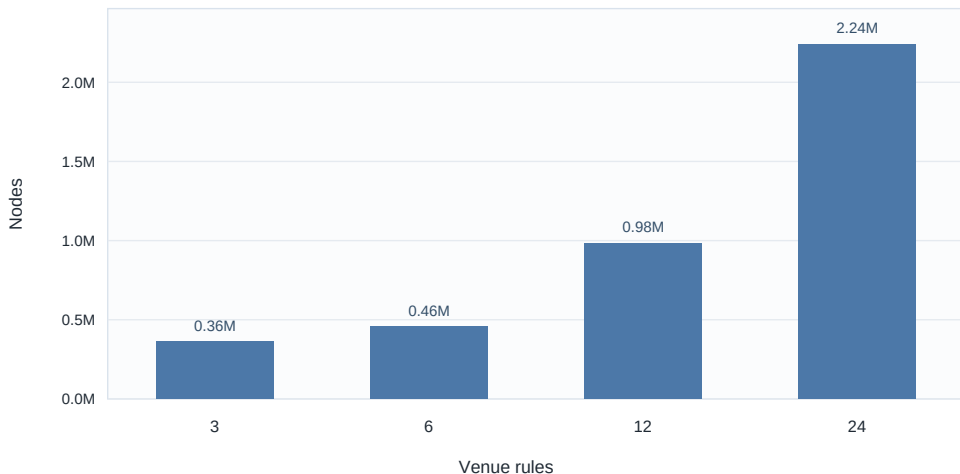
Live Demo of Benchmark

(Demo)

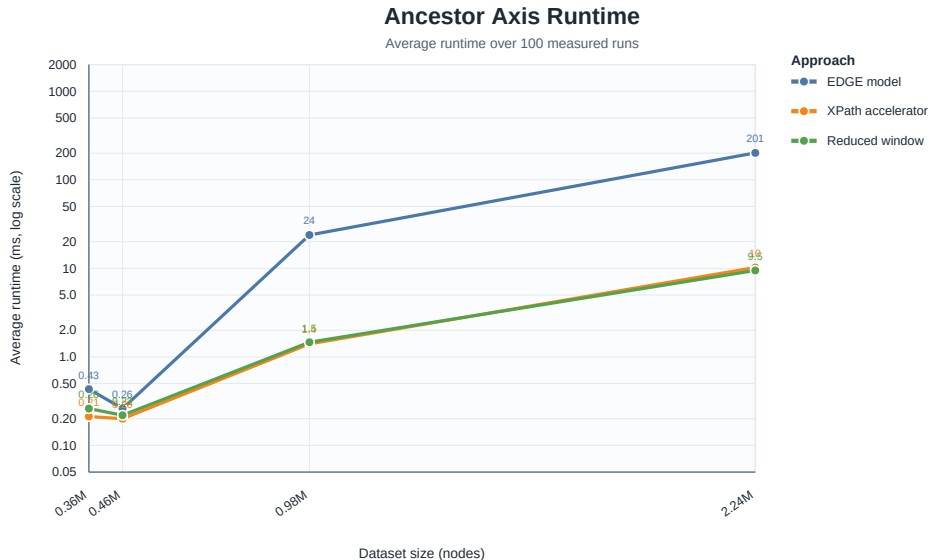
Benchmark: Dataset Size

Benchmark Dataset Sizes

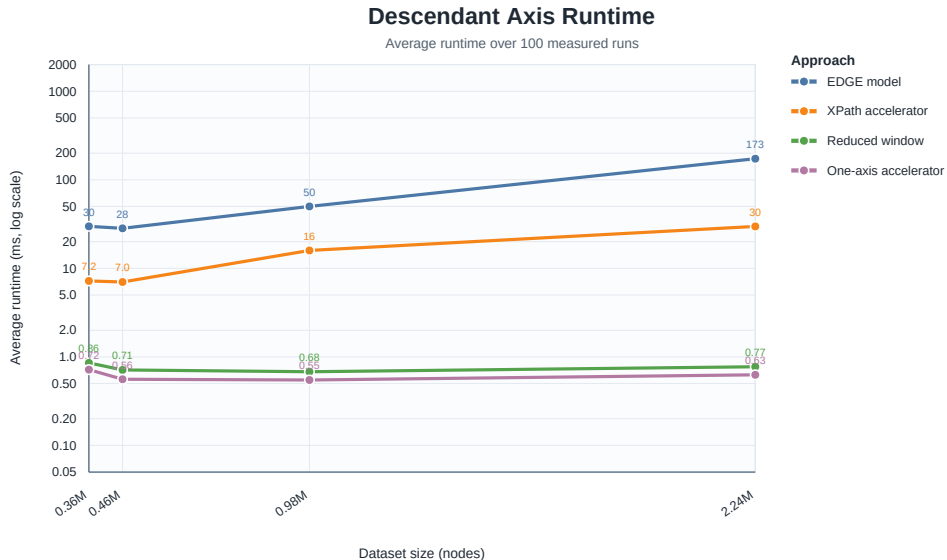
Parsed DBLP subset size by number of venue rules



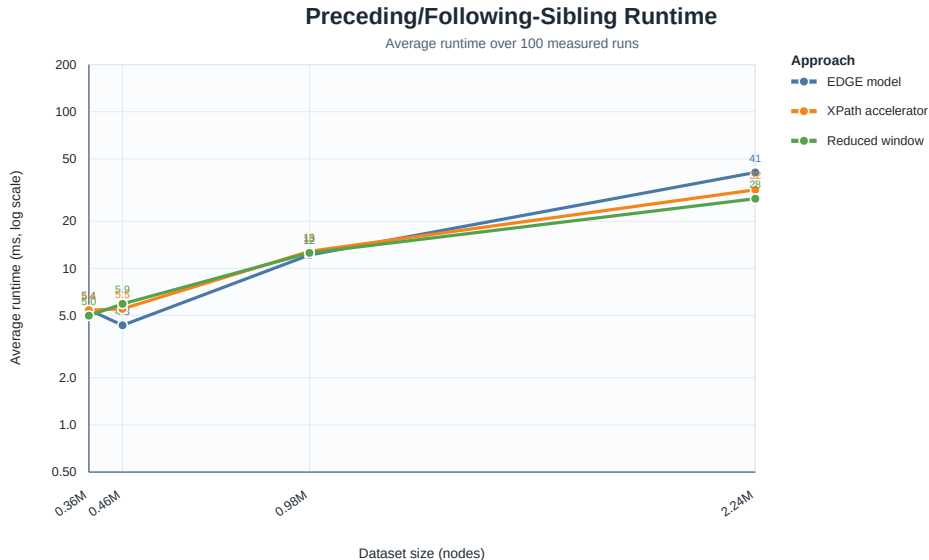
Benchmark Results: Ancestors



Benchmark Results: Descendants



Benchmark Results: Siblings



What went well, what not?

- What went well
 - Implementing a SAX handler was not that hard, due to linked tutorial
 - Working with the tree data structures was not as difficult as expected
- What was new?
 - SAX
 - Recursive SQL
- Difficulties
 - Initially difficulties with getting the SAX parser to accept/parse the XML correctly
 - Also had some out-of-memory issues
 - We also at first did not understand how to implement the windows approach